ANALYSIS ON WATERSHED PROJECTS IN INDIA



SEMESTER IV

M.SC. COMPUTER SCIENCE (DATA ANALYTICS)

(2023-2025)

SUBMITTED BY

AKHILESH SINGH YADAV

23203002

UNDER THE GUIDANCE OF

DR. KEERTHY A.S.

MARCH 2025



DEPARTMENT OF COMPUTER SCIENCE

RAJAGIRI COLLEGE OF SOCIAL SCIENCES (AUTONOMOUS)

KALAMASSERY- 683104

CERTIFICATE

*This is to certify that the project titled "**Analysis on Watershed Projects in India**" is a bona fide work carried out by **Akhilesh Singh Yadav (Reg No: 23203002)** in partial fulfillment of the requirements for the award of the Master's degree in Computer Science (Data Analytics) of Rajagiri College of Social Sciences (Autonomous), affiliated to Mahatma Gandhi University, during the year 2023- 2025. This project report has been approved as it satisfies the academic requirement of project work prescribed for the M.Sc. Computer Science (Data Analytics).*

Dr. Keerthy A.S.
Project Guide

Dr. Bindiya M Varghese
Dean- Computer Science

Examiner –I

Examiner -II

Place :

Date :

DECLARATION

I hereby declare that this project work and the report submitted to the Department of Computer Science, Rajagiri College of Social Sciences (Autonomous), Kalamassery in partial fulfilment of the award of the degree of M.Sc. Computer Science (Data Analytics) is an outcome of my own work.

A copy of the report has been submitted to the organization for which this project was developed.

To the best of my knowledge, this Project work or parts there does not form a part of any other project work or thesis on the basis of which a degree or award was conferred on an earlier occasion.

Date :

Place : Akhilesh Singh Yadav

ACKNOWLEDGEMENT

I consider myself a very lucky individual as I was provided with a great opportunity to be a part of the company **KVJ Analytics** to work as an **Intern**. I am also grateful for having a chance to meet so many wonderful people and professionals who led me through this internship period.

I express my deepest thanks to **Mr. Jomon Jospeh, C.E.O The Strategist** for his valuable guidance, advice, and encouragement at various stages of my internship period. I choose this moment to acknowledge his contribution gratefully, which was extremely valuable for my study both theoretically and practically.

I would like to express my sincere thanks to **Fr. (Dr.) Saju MD CMI, Principal, Rajagiri College of Social Sciences (Autonomous),** and **Dr. Bindiya M Varghese, Dean, Department of Computer Science,** for their support in the successful completion of this internship work. I acknowledge with gratitude the kind patronage, loving inspiration, and timely guidance I have received from my project guide and project coordinator, **Dr. Keerthy A.S**. I would like to thank all the college faculty members and management.

Akhilesh Singh Yadav

ABSTRACT

India’s watershed projects play a crucial role in water conservation, soil management, and agricultural sustainability. Despite substantial government investments, questions remain about their efficiency and actual impact on rural communities. This study thoroughly analyses watershed projects by looking at social variables including approved budgets, treasury money releases, and spending trends in addition to financial data such as both the number of farmers and the creation of jobs benefited.

This study employs data analytics tools to examine whether these programs meet their stated socio-economic goals, identify inefficiencies in fund utilization, and reveal discrepancies in resource allocation. The study identifies clear trends in project execution and financial management by classifying states according to cash flow efficiency. Predictive modeling also helps with improved financial planning by estimating costs per hectare.

The study's conclusions provide information on how watershed projects can be best utilized to guarantee efficient resource distribution and noticeable advantages for farming communities. The findings are intended to help authorities improve future initiatives, increase openness, and guarantee sustainable rural development.

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CHAPTER 1
INTRODUCTION

1.1 Introduction

In a nation like India, where water shortages and sustainable agriculture are still top-of-mind issues, watershed projects are an essential tool for preserving water, preserving soil, and sustaining rural economies. Watershed projects are created to make farming more resilient and sustainable in the long term, but their success hinges greatly on the effectiveness with which resources are distributed and managed.

This research, Analysis on Watershed Projects in India, assesses the fiscal effectiveness and socio-economic value of these initiatives. Through data analytics, principal points like fund allocation, expenditures incurred, number of farmers benefited, and employment generated (in mandays) are reviewed. The objective is to establish the extent to which these projects convert investments into concrete results for rural communities.

Through in-depth analysis and graphical illustrations, this study illuminates the efficacy of watershed management at the policy and implementation levels. By filling the gap between strategic planning and on-ground implementation, the study hopes to contribute to more effective water conservation, enhanced agricultural productivity, and improved rural livelihoods.

1.2 Organization Profile

KVJ Analytics, based in Kochi, is a top Data Analytics & Visualization solutions provider, focusing on turning raw data into business-driven insights. They apply cutting-edge analytics across various domains, such as Customer Analytics, Marketing Analytics, Sales Analytics, Performance Analytics, HR Analytics, and E-commerce Analytics. They also specialize in Outsourced Data Analysis and Data Visualization, enabling organizations to make informed decisions based on data, which improves performance and fuels growth. Through the use of advanced tools and techniques, they enable companies to derive meaningful insights, streamline processes, and enhance decision-making.

1.3 Objectives of Analysis on Watershed Projects in India

- Beneficiary Assessment of Watershed Projects: Analyzing Budget, Execution, Mandays Generated and Farmers Benefitted.
- Fund Flow Analysis and Efficiency of Watershed Projects in India.
- Predicting Cost Per Hectare for Watershed Projects: Forecasting Funding Efficiency Based on Location and Investment.

CHAPTER 2
DATASET AND DATA PREPROCESSING

2.1 Problem Statement

In India, watershed management initiatives are crucial for enhancing rural livelihoods, preserving water, and bolstering agriculture. Nevertheless, these programs frequently suffer from ineffective funding distribution, implementation hold-ups, and unclear advantages for farmers. There are still issues with money utilization transparency despite large government investments. Unpredictable financing trends also cause problems, which slow down project execution and postpone vital assistance for rural populations.

In order to optimize resource allocation, this study analyses the operational and financial dynamics of watershed projects, assesses their effects on beneficiaries, and creates a predictive model. The study intends to improve decision-making procedures by utilizing data-driven insights, guaranteeing that these initiatives are carried out more effectively, openly, and with more advantages for farmers and rural communities.

2.2 About Dataset

The data range over 4 csv files with multiple instances and different attributes denoting the information of watershed projects, such as the budgeted amount, treasury released, expenditure incurred, farmers benefited, and mandays created. Table 2.2.1 contains details of sanctioned projects, Table 2.2.2 displaying details of mandays generated and farmers benefitted, Table 2.2.3 contains details of total expenditure and Table 2.2.4 displays details of treasury receipt.

2.2.1 Dataset: Details of Sanctioned Projects (1150 instances, 9 attributes)

Field	Description
State Name	Name of the state where the project is implemented
District Name	Name of the district within the state
Financial Year	The fiscal year in which the project was sanctioned.
Project Name	The official name of the watershed project
Project Sanctioned Date	The date on which the project was approved

Area Sanctioned	The total area (in hectares) covered under the project
Total Cost	The total estimated cost of the project
Central Share	The portion of the total cost funded by the central government.
State Share	The portion of the total cost funded by the state government

Table 2.2.1 Details of Sanctioned Projects

2.2.2 Dataset: State Wise Mandays Generated and Farmers Benefitted (30 instances, 11 attributes)

Field	Description
State Name	Name of the state
No. Of Mandays Generated (in mandays)	Total workdays created
SC	Total SC farmers benefitted
ST	Total ST farmers benefitted
Others	Other farmers benefitted
Total	SC+ST+Others
Female	Total female farmers benefitted
Small farmers	Total small farmers benefitted
Marginal farmers	Total marginal farmers benefitted
Landless	Total landless farmers benefitted
BPL	Total below poverty line farmers benefitted

Table 2.2.2 State Wise Mandays Generated and Farmers Benefitted

2.2.3 Dataset: State Wise Total Expenditure (30 instances, 9 attributes)

Field	Description
State Name	Name of the state
District expenditure	Funds spent at the district level
Block expenditure	Funds spent at the block level
Gram expenditure	Funds spent at the gram panchayat level
Village expenditure	Funds spent at the village level
Total transactions	Count of all financial transactions
Transactions mapped with projects	Transactions linked to specific projects
Transactions mapped with work id	Transactions linked to specific work IDs
Total expenditure	SLNA utilizes funds

Table 2.2.3 State Wise Total Expenditure

2.2.4 Dataset: State Wise Treasury Receipt (30 instances, 4 attributes)

Field	Description
State Name	Name of the state
Financial Year	Year of fund release
State's share Release From Treasury	State Share Release from State Treasury to Slna
Central's share Release From Treasury	Central Share Release from State Treasury to Slna

Table 2.2.4 State Wise Treasury Receipt

2.3 Source of Dataset

Watershed Development Component of PMKSY 2.0 (Pradhan Mantri Krishi Sinchayee Yojana)-Management Information System (MIS)-Department of Land Resources-Ministry of Rural Development-Government of India.

2.4 Data Preprocessing

All pre-processing steps were carried out using Python.

- Merged 30 (state-wise) Excel files into one: Combined state-wise data files into a single dataset for streamlined analysis.
- Converted Project Sanction Date to datetime format: Changed the format for easier manipulation and analysis.
- Added a new column, "Total Expenditure": Calculated the sum of expenditures from various columns.
- Scaled Total Expenditure to Lakhs: Scaled down values for better readability.
- Converted Total Expenditure to Lakhs: Scaled down values for better readability.
- Grouped treasury receipt data by state: Aggregated values to get total receipts per state.
- Converted Total Treasury Receipt to Lakhs: Ensured consistency with expenditure data.
- Generated total projects done in each state: Counted the total number of projects per state.
- Calculated total farmers benefitted: Summed up various farmer categories (SC, ST, Others, female, BPL, marginal, landless, small) to get a comprehensive impact measure.
- Aggregated datasets: Total Sanctioned Budget, Total Treasury & Total Expenditure in a master dataset for clustering.
- Calculated new efficiency metrices using the formulas given in Figure 2.4.1

$$\text{Release Efficiency} = \left(\frac{\text{Total Release From Treasury}}{\text{Total Sanctioned Amount}}\right) \times 100$$

$$\text{Expenditure Efficiency} = \left(\frac{\text{Total Expenditure}}{\text{Total Release From Treasury}}\right) \times 100$$

$$\text{Overall Utilization} = \left(\frac{\text{Total Expenditure}}{\text{Total Sanctioned Amount}}\right) \times 100$$

Figure 2.4. 1 Efficiency Metrices

CHAPTER 3
DATA MODELING AND ANALYSIS

3.1 Data Exploration and Analysis

Dataset: Details of Sanctioned Projects

Figure 3.1.1 shows the first five records of the dataset along with their respective datatypes.

```
First 5 Rows:
   S.No.      State Name          District Name Financial Year  \
0      1  ANDHRA PRADESH  Alluri Sitharama Raju        2021-22
1      2  ANDHRA PRADESH             Anakapalli        2021-22
2      3  ANDHRA PRADESH             Anakapalli        2021-22
3      4  ANDHRA PRADESH              ANANTAPUR        2021-22
4      5  ANDHRA PRADESH              ANANTAPUR        2021-22

                          Project Name Project Sanction Date  \
0  Alluri Sitharama Raju-WDC - 1 /2021-22            20-01-2022
1            Anakapalli-WDC - 1 /2021-22            20-01-2022
2            Anakapalli-WDC - 2 /2021-22            20-01-2022
3             ANANTAPUR-WDC - 1 /2021-22            20-01-2022
4             ANANTAPUR-WDC - 2 /2021-22            20-01-2022

   Area Sanctioned  Total Cost  Central Share  State Share
0             5000     1100.00         660.00       440.00
1             5000     1100.00         660.00       440.00
2             5000     1100.00         660.00       440.00
3             3460      968.80         581.28       387.52
4             3199      895.72         537.43       358.28
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1150 entries, 0 to 1149
Data columns (total 10 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   S.No.                  1150 non-null   int64
 1   State Name             1150 non-null   object
 2   District Name          1150 non-null   object
 3   Financial Year         1150 non-null   object
 4   Project Name           1150 non-null   object
 5   Project Sanction Date  1150 non-null   object
 6   Area Sanctioned        1150 non-null   int64
 7   Total Cost             1150 non-null   float64
 8   Central Share          1150 non-null   float64
 9   State Share            1150 non-null   float64
dtypes: float64(3), int64(2), object(5)
memory usage: 90.0+ KB
None
```

Figure 3.1. 1 Details of Sanctioned Projects

Dataset: State Wise Mandays Generated and Farmers Benefitted

Figure 3.1.2 shows the first five records of the dataset along with their respective datatypes.

```
First 5 Rows:
   S.No.         State Name  No. of Mandays generated (in mandays)     SC  \
0      1     ANDHRA PRADESH                                 224100   2471
1      2  ARUNACHAL PRADESH                                 318234      0
2      3              ASSAM                                2019053  17027
3      4              BIHAR                                 160346  12979
4      5       CHHATTISGARH                                 390697   2150

      ST  Others   Total  Female  Small Farmers  Marginal Farmers  Landless  \
0   1410   10016   13897    2922           7958              5106       308
1   9465       0    9465    3824           2429              2001       748
2  47842   68984  133853   23756          47340             37979     26372
3   5921   37901   56801   15308          10510             11351     13304
4   6365    3906   12421    3395           6197              4110       883

     BPL
0   1323
1   3976
2  45748
3  31133
4   8878
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 12 columns):
 #   Column                              Non-Null Count  Dtype
---  ------                              --------------  -----
 0   S.No.                               30 non-null     int64
 1   State Name                          30 non-null     object
 2   No. of Mandays generated (in mandays)  30 non-null     int64
 3   SC                                  30 non-null     int64
 4   ST                                  30 non-null     int64
 5   Others                              30 non-null     int64
 6   Total                               30 non-null     int64
 7   Female                              30 non-null     int64
 8   Small Farmers                       30 non-null     int64
 9   Marginal Farmers                    30 non-null     int64
 10  Landless                            30 non-null     int64
 11  BPL                                 30 non-null     int64
dtypes: int64(11), object(1)
memory usage: 2.9+ KB
None
```

Figure 3.1. 2 State Wise Mandays Generated

Dataset: State Wise Total Expenditure

Figure 3.1.3 shows the first five records of the dataset along with their respective datatypes.

```
First 5 Rows:
    S.No.         State Name  District Expenditure  Block Expenditure  \
0    1.0     ANDHRA PRADESH          2.644203e+09        165901681.0
1    2.0  ARUNACHAL PRADESH          1.379041e+09        784182250.0
2    3.0              ASSAM          1.895125e+08         10623900.0
3    4.0              BIHAR          1.634283e+09        100061308.0
4    5.0       CHHATTISGARH          7.118831e+08        704907665.7

   Gram Expenditure  Village Expenditure  Total Transactions  \
0               0.0         8.801015e+06             11639.0
1        21945000.0         5.632356e+08              2107.0
2       108201134.0         9.120019e+08             23566.0
3         8526993.0         8.677215e+07             20274.0
4         6929571.0         1.062943e+09             56898.0

   Transactions Mapped with Project  Transactions Mapped with Work-Id\t  \
0                               1.0                                0.0
1                               1.0                                0.0
2                           15602.0                                0.0
3                            3042.0                                0.0
4                           11736.0                                0.0

   Total Expenditure  Total Expenditure (in La)
0       2.818906e+09                28189.060530
1       2.748404e+09                27484.042460
2       1.220339e+09                12203.394166
3       1.829644e+09                18296.435140
4       2.486663e+09                24866.632681
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 37 entries, 0 to 36
Data columns (total 11 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   S.No.                           30 non-null     float64
 1   State Name                      30 non-null     object
 2   District Expenditure            30 non-null     float64
 3   Block Expenditure               30 non-null     float64
 4   Gram Expenditure                30 non-null     float64
 5   Village Expenditure             30 non-null     float64
 6   Total Transactions              30 non-null     float64
 7   Transactions Mapped with Project  30 non-null     float64
 8   Transactions Mapped with Work-Id    30 non-null     float64
 9   Total Expenditure               30 non-null     float64
 10  Total Expenditure (in La)       30 non-null     float64
dtypes: float64(10), object(1)
memory usage: 3.3+ KB
None
```

Figure 3.1. 3 State Wise Total Expenditure

Dataset: State Wise Treasury Receipt

Figure 3.1.4 shows the first five records of the dataset along with their respective datatypes.

```
First 5 Rows:
    S.No.         State Name Financial Year  Treasury Reciept  Central Share  \
0      1     ANDHRA PRADESH      2022-2023         862275000      517365000
1      2     ANDHRA PRADESH      2021-2022         146337656              0
2      3  ARUNACHAL PRADESH      2021-2022          64444000              0
3      4  ARUNACHAL PRADESH      2022-2023        1034028000      930616000
4      5              ASSAM      2022-2023         607720888      540200000

   State Share
0    344910000
1    146337656
2            0
3    103412000
4     67520888
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 52 entries, 0 to 51
Data columns (total 6 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   S.No.             52 non-null     int64
 1   State Name        52 non-null     object
 2   Financial Year    52 non-null     object
 3   Treasury Reciept  52 non-null     int64
 4   Central Share     52 non-null     int64
 5   State Share       52 non-null     int64
dtypes: int64(4), object(2)
memory usage: 2.6+ KB
None
```

Figure 3.1. 4 Sate Wise Treasury Receipt

3.2 Data Modeling

In this study, two key machine learning techniques—K-Means Clustering **and** Random Forest Regression—were employed to analyze watershed projects in India. These methods helped categorize states based on fund utilization efficiency and predict the estimated **cost per hectare** for new projects, respectively.

3.2.1 Clustering States Using K-Means

In order to evaluate the effectiveness of fund usage in various states, K-Means Clustering as depicted in Figure 3.2.1 was used to classify them according to important efficiency criteria like:

- **Release Efficiency** – Percentage of sanctioned funds actually released.
- **Expenditure Efficiency** – Percentage of released funds utilized.
- **Overall Utilization** – How effectively the allocated funds were spent on projects.
-

```
# Applying K-Means clustering (k=3)
optimal_k = 3
kmeans = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(features_scaled)

# Display clustered data
df[["State_Name", "Release_Efficiency", "Expenditure_Efficiency", "Overall_Utilization", "Cluster"]].head()
```

Figure 3.2. 1 Clustering Model

3.2.2 Predicting Cost Per Hectare Using Random Forest Regressor

The cost per hectare for new watershed projects was predicted using a Random Forest Regressor as depicted in Figure 3.2.2. When training this model on historical data, characteristics like:

- Total Sanctioned Cost
- Central and State Share Contributions
- Area Sanctioned for the Project
- Efficiency Ratios of Fund Utilization
- State and District-wise Historical Averages

```
# Prepare features and target
features = ['Central Share', 'State Share', 'Central_Funding_Ratio', 'State_Funding_Ratio',
            'State_Avg_Area', 'District_Avg_Cost', 'Area_Central_Interaction',
            'Central_State_Ratio', 'Log_Central_Share',
            'State_Name_Encoded', 'District_Name_Encoded']
X = data[features]
y = data['Cost_Per_Hectare']
```

```
# Train and evaluate model
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
rmse = math.sqrt(mse)
r2 = r2_score(y_test, y_pred)
print(f'Mean Squared Error (MSE): {mse:.6f}')
print(f'Root Mean Squared Error (RMSE): {rmse:.6f}')
print(f'R-squared (R²): {r2:.4f}')
```

Figure 3.2. 2 Predictive Model

3.3 Deployment and Optimization

Deployment Strategy

Given that the project was created on Google Colab, models were implemented to be executed in a cloud-based platform to make them accessible and easily executable. Outcomes such as clustering insights and cost estimates may be exported as reports or incorporated into decision-making for policymakers and stakeholders.

- Cloud-Based Execution: Running the models in Google Colab ensures seamless access to datasets and computational resources without requiring local installations.
- Data Export and Sharing: The clustering results and cost predictions can be saved as CSV files or visualized through graphs for further analysis.
- Scalability: The models can process new data over time, making it possible to retrain and improve predictions based on updated information.

Optimization Techniques

The following steps were taken in order to increase model accuracy:

1. **Hyperparameter Tuning:**

- For K-Means Clustering, the Elbow Method was employed to select the ideal number of clusters, with the optimal grouping of states in terms of efficiency.
- For the Random Forest Regressor, various hyperparameters (e.g., the trees number and tree depth) were optimized with Grid Search Cross-Validation to enhance the model's predictive performance.

2. **Feature Selection and Engineering:**

- Further applicable characteristics were developed, like funding efficiency ratios and past cost trends, to better model precision.
- Numerical values were normalized and scaled as required to provide uniformity among various financial indicators.

3. **Model Performance Evaluation:**

- The Random Forest model with an R^2 score of 0.88 indicated a strong prediction accuracy.

CHAPTER 4
OBJECTIVE 1

4.1 Discussion

Objective: Beneficiary Assessment of Watershed Projects: Analyzing Budget, Execution, Mandays Generated and Farmers Benefitted.

VS Code was used for data pre-treatment, Python Collab was used for data manipulation and visualization, and Power BI was used to build geographical maps. The data could be handled and visualized well because to this procedure.

4.2 Implementation

To assess the impact of watershed projects, data on budget allocation, fund utilization, mandays generated, and farmers benefitted was analyzed. Summary statistics, correlation analysis, and visualizations were used to identify trends and inefficiencies. This helped in understanding how effectively funds translated into employment and agricultural benefits for rural communities.

State Wise Total Budget Distribution

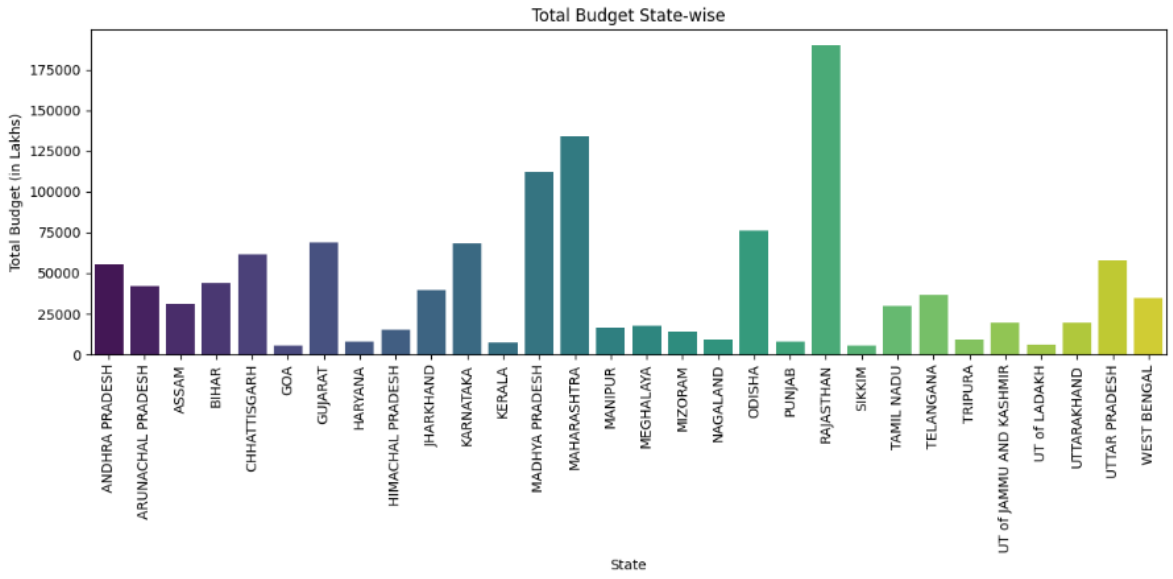


Figure 4.2. 1 Total Budget State Wise

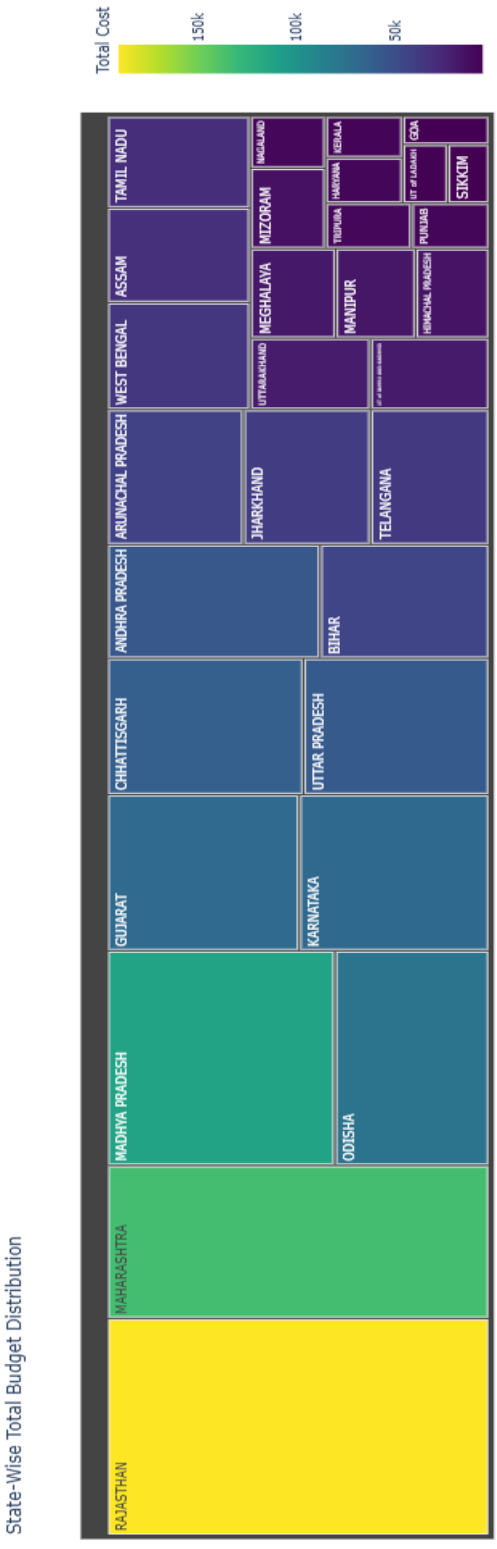


Figure 4.2. 2 Total Budget State Wise-Tree Map

Key Inferences from Figure 4.2.1 and Figure 4.2.2

- Rajasthan has the highest budget allocation among all states, significantly surpassing others, indicating a major focus on development or expenditure in this state.
- Maharashtra and Punjab also have substantial budgets, reflecting their economic significance and potential investment in infrastructure, welfare, or public services.
- Some northeastern states and union territories have comparatively lower budgets, which may indicate either lower population sizes, lesser economic activity, or different funding priorities.

State Wise Total Projects Executed

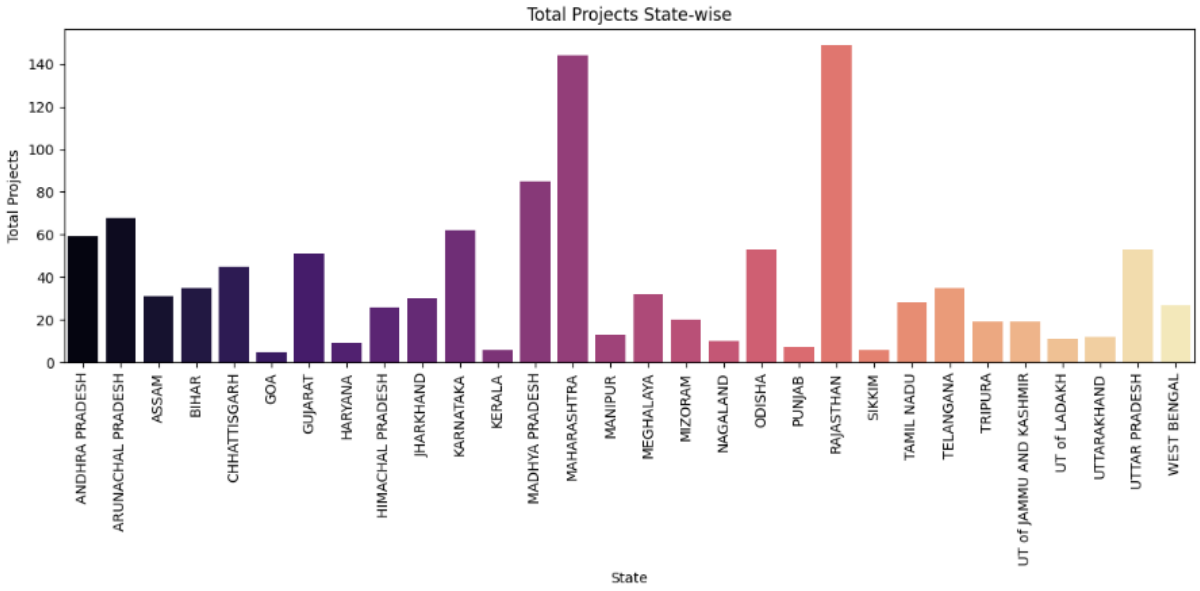


Figure 4.2. 3 Total Projects State Wise

Key Inferences from Figure 4.2.3

- Maharashtra and Rajasthan have the highest number of total projects compared to other states, indicating strong project participation or allocation in these regions.
- Some states like Goa, Sikkim, and the Union Territories have significantly fewer projects, suggesting either lower participation, fewer initiatives, or limited resources in these areas.

State Wise Mandays Generated

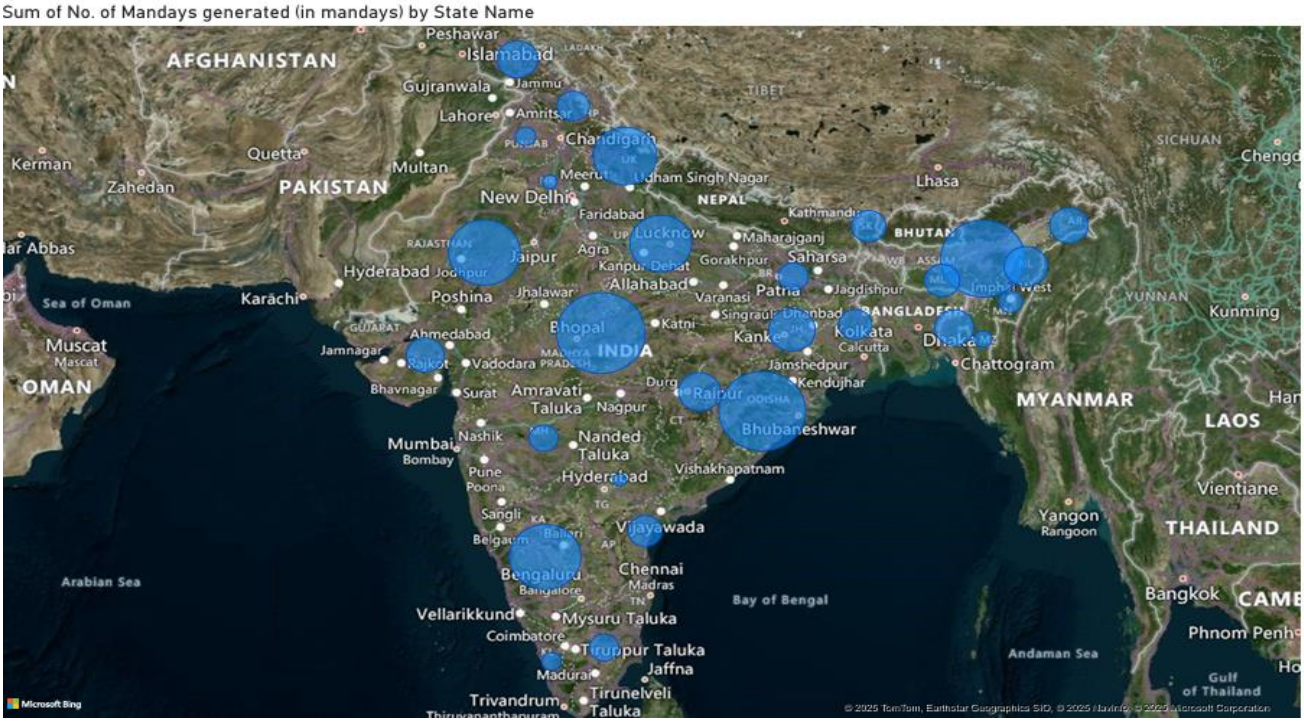


Figure 4.2. 4 State Wise Mandays Generated-BI Map

Key Inferences from Figure 4.2.4

- States like Bihar, Maharashtra, Odisha, and Punjab have larger bubbles, indicating they generated the highest number of mandays compared to other states. This suggests higher employment or work allocation in these regions.
- States like Goa, Himachal Pradesh, and some northeastern states have much smaller bubbles, implying relatively fewer mandays generated, which could be due to lower workforce participation, fewer projects, or limited funding.

State Wise Farmers Benefitted (Breakdown by Categories)

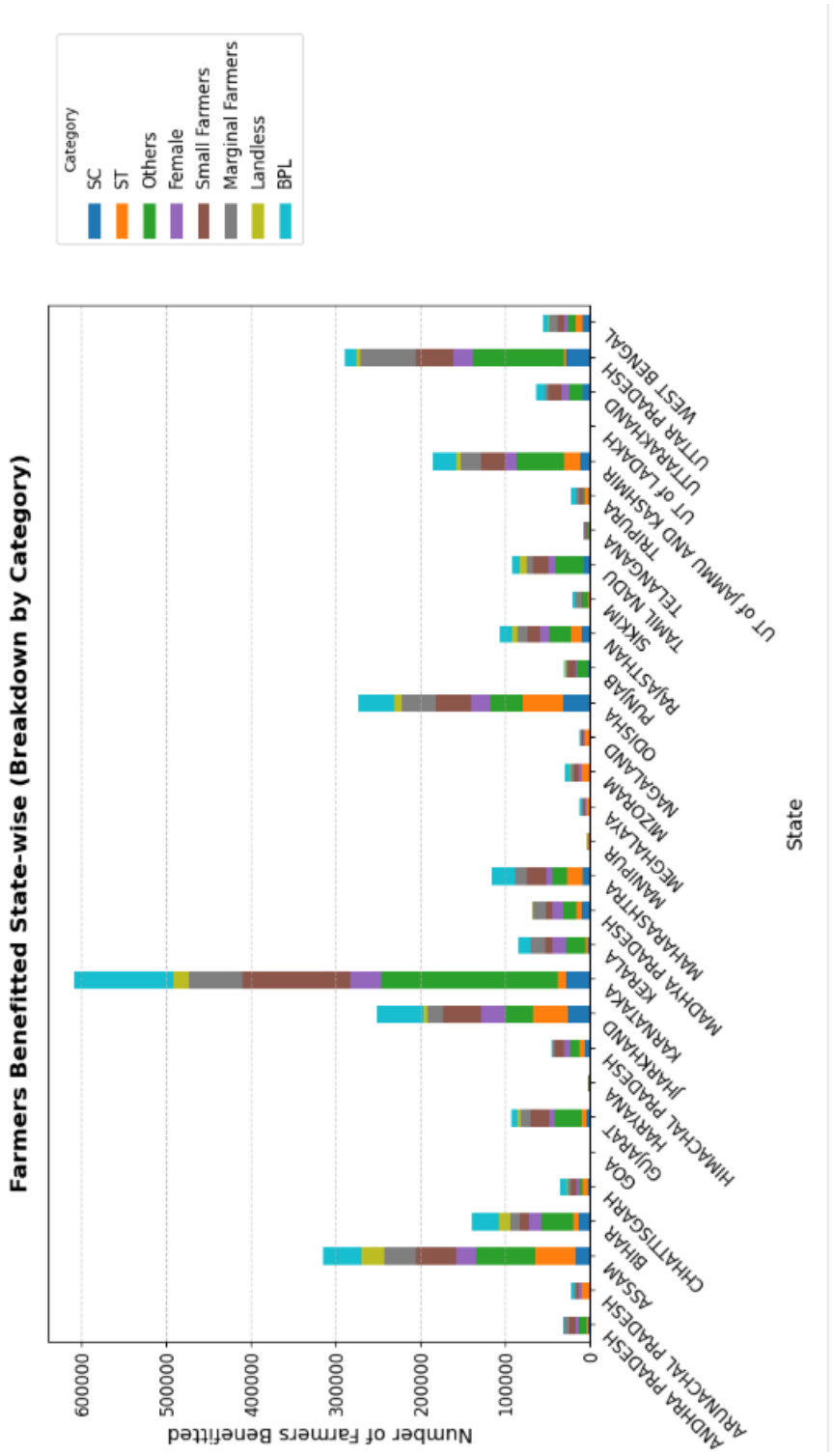


Figure 4.2. 5 Farmers Benefitted

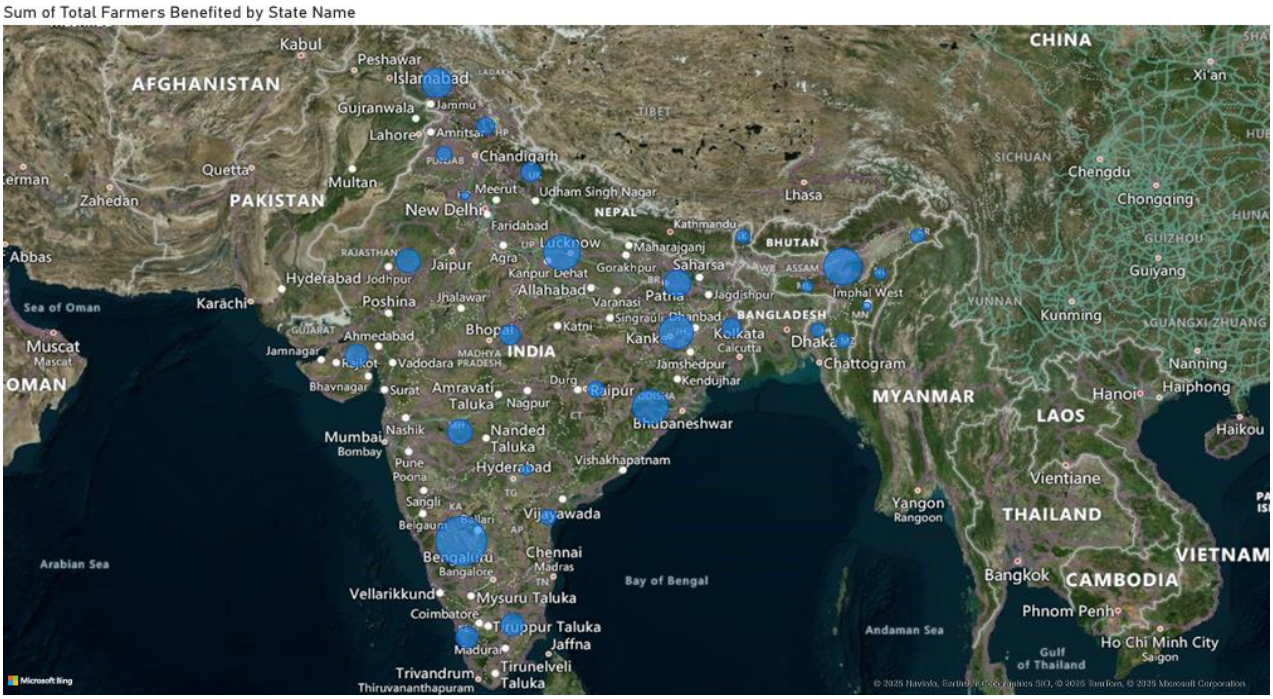


Figure 4.2. 6 Farmers benefitted-BI Map

Key Inferences from Figure 4.2.5 and Figure 4.2.6

- Madhya Pradesh has the highest number of farmers benefitted**,** followed by Bihar and Maharashtra. This indicates that these states have significant agricultural support programs reaching a large number of farmers.
- The "Small Farmers" and "BPL" categories have a substantial share across most states**,** suggesting that government schemes focus more on economically weaker sections.
- Some states like Goa, Sikkim, and the Union Territories have minimal farmer benefits, possibly due to smaller agricultural populations or limited government schemes in these regions.
- The "Female" category is consistently represented across states, but in smaller numbers compared to others**,** indicating that while women farmers receive benefits, they still form a minority in agricultural assistance programs.

4.3 Results

Correlation analysis of Total Cost, Total Projects, No.of Mandays Generated and Total Farmers Benefitted

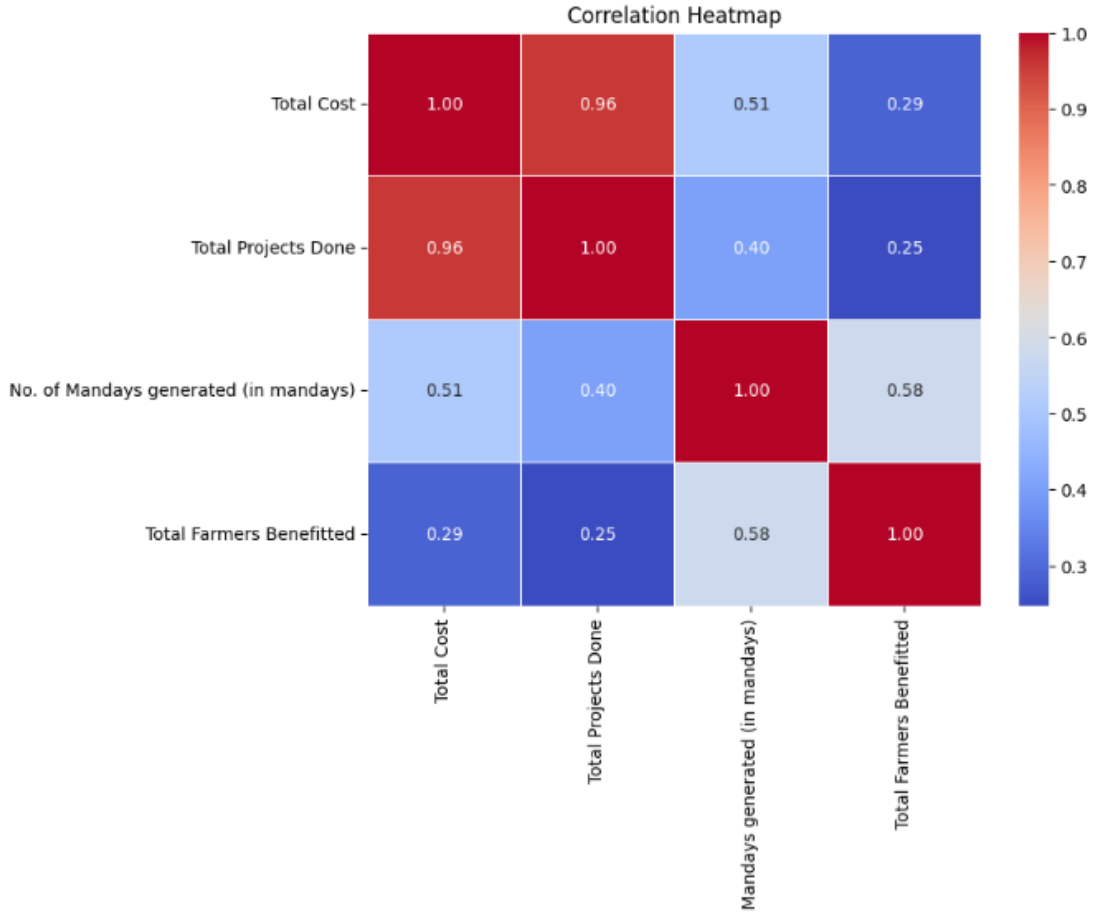


Figure 4.3. 1 Correlation Heatmap

Insights from Figure 4.3.1

- **Strong Positive Correlation between Total Cost and Total Projects Done**

The correlation coefficient of 0.96 indicates that as the number of projects increases, the total cost increases proportionally. This suggests that project costs scale directly with the number of projects undertaken.

- **Moderate Correlation Between Total Cost and Mandays Generated**

With a correlation coefficient of 0.51, there's a moderate link between total project costs and

the number of mandays generated. This suggests that higher costs tend to create more employment opportunities, but other factors also play a role.

- **Weak Correlation Between Total Cost and Farmers Benefitted**

A correlation of 0.29 indicates a weak connection between spending and the number of farmers benefitted. This means higher costs don't necessarily translate to significantly better outcomes for farmers.

- **Moderate Positive Correlation Between Mandays Generated and Farmers Benefitted**

A correlation of 0.58 suggests that labor-intensive projects tend to involve and benefit more farmers. More employment opportunities generally lead to a higher number of farmers gaining support.

- **Strong Correlation Between Total Projects Done and Total Cost**

The perfect correlation of 1.0 confirms a direct proportionality between the number of projects and total costs, indicating consistent budgeting and planning.

- **Weak Correlation Between Total Projects Done and Farmers Benefitted**

With a correlation of 0.25, the number of projects completed does not strongly impact the number of farmers benefitted. This could indicate inefficiencies or a misalignment between project execution and actual farmer support.

- **Moderate Correlation Between Mandays Generated and Total Projects Done**

A correlation of 0.40 shows that while more projects generally lead to more employment, the increase in mandays is not strictly proportional to the number of projects.

Inferences

Smarter Fund Allocation: Despite significant spending and multiple projects, the weak link between costs and the number of farmers benefitted suggests that funds may not always be used efficiently. Improving project design and allocation could help maximize impact.

Prioritizing Labor-Intensive Projects: Since labor-intensive projects show a stronger connection to farmer benefits, focusing on these could lead to better community outcomes and more direct support for farmers.

Assessing Project Effectiveness: The weak correlation between spending and farmer benefits highlights the need for better evaluation. Ensuring that investments lead to real improvements for farmers should be a key priority.

Maximizing Impact Through Employment: Projects that generate more mandays tend to benefit more farmers, suggesting that prioritizing employment-heavy initiatives could enhance social and economic impact.

CHAPTER 5
OBJECTIVE 2

5.1 Discussion

Objective: Fund Flow Efficiency Analysis of Watershed Projects in India.

The analysis of fund flow provided a clearer picture of how allocated funds were distributed and utilized in watershed projects. It revealed delays and inefficiencies in fund release and spending, which can impact project execution. Understanding these patterns is essential for improving financial planning and ensuring that resources reach communities efficiently and on time.

Steps in Fund Flow of Watershed projects

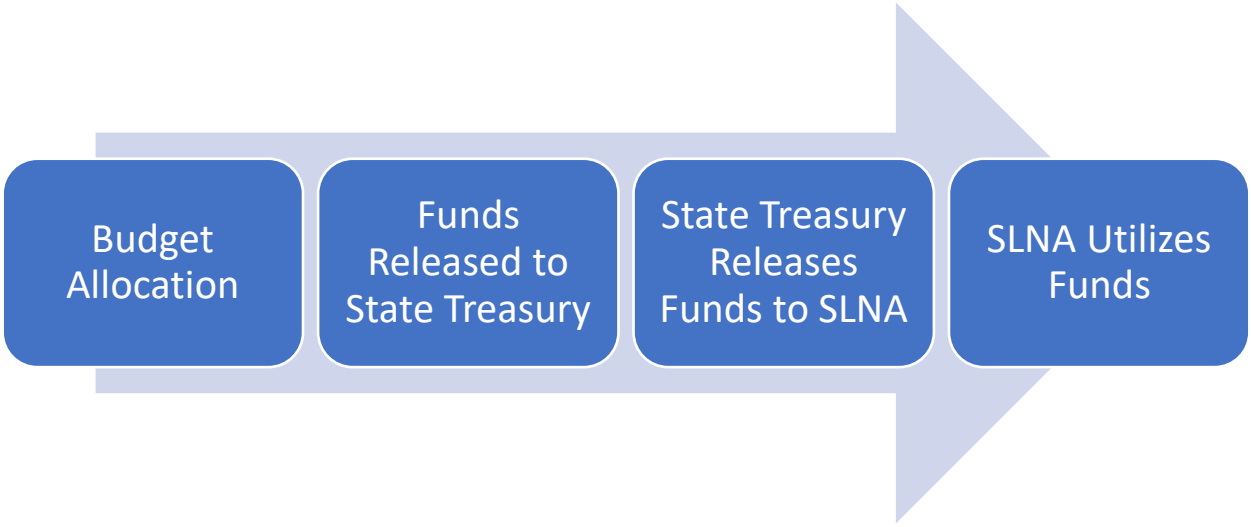


Figure 5.1. 1 Steps in Fund Flow

5.2 Implementation

Aggregated Datasets: Total Sanctioned Budget, Total Treasury & Total Expenditure in a master dataset for clustering as given in Figure 5.2.1

	State_Name	Total_Sanctioned_Amount	Total_Release_From_Treasury	Total_Expenditure
0	ANDHRA PRADESH	55531.30	8622.75000	5853.57455
1	ARUNACHAL PRADESH	42256.20	10340.28000	6214.27380
2	ASSAM	31060.26	6077.20888	4399.29102
3	BIHAR	44097.18	16155.00000	3677.25264
4	CHHATTISGARH	61366.19	7870.74048	6714.18000

Figure 5.2. 1 Aggregated Dataset

Calculating new efficiency columns using the following efficiency formulas:

New efficiency columns were calculated from the formulas shown in figure 5.2.2 and the result columns depicted in Figure 5.2.3

$$\text{Release Efficiency} = \left(\frac{\text{Total Release From Treasury}}{\text{Total Sanctioned Amount}}\right) \times 100$$

$$\text{Expenditure Efficiency} = \left(\frac{\text{Total Expenditure}}{\text{Total Release From Treasury}}\right) \times 100$$

$$\text{Overall Utilization} = \left(\frac{\text{Total Expenditure}}{\text{Total Sanctioned Amount}}\right) \times 100$$

Figure 5.2. 2 Efficiency Formulas

	State_Name	Total_Sanctioned_Amount	Total_Release_From_Treasury	Total_Expenditure	Release_Efficiency	Expenditure_Efficiency	Overall_Utilization
0	ANDHRA PRADESH	55531.30	8622.75000	5853.57455	15.527729	67.885240	10.541036
1	ARUNACHAL PRADESH	42256.20	10340.28000	6214.27380	24.470445	60.097732	14.706182
2	ASSAM	31060.26	6077.20888	4399.29102	19.565866	72.389992	14.163729
3	BIHAR	44097.18	16155.00000	3677.25264	36.634996	22.762319	8.338975
4	CHHATTISGARH	61366.19	7870.74048	6714.18000	12.825858	85.305570	10.941171

Figure 5.2. 3 Calculated Efficiency Metrices

5.3 Results

Figure 5.3.1 displays a side-by-side bar chart for comparing Treasury Release vs. Expenditure;

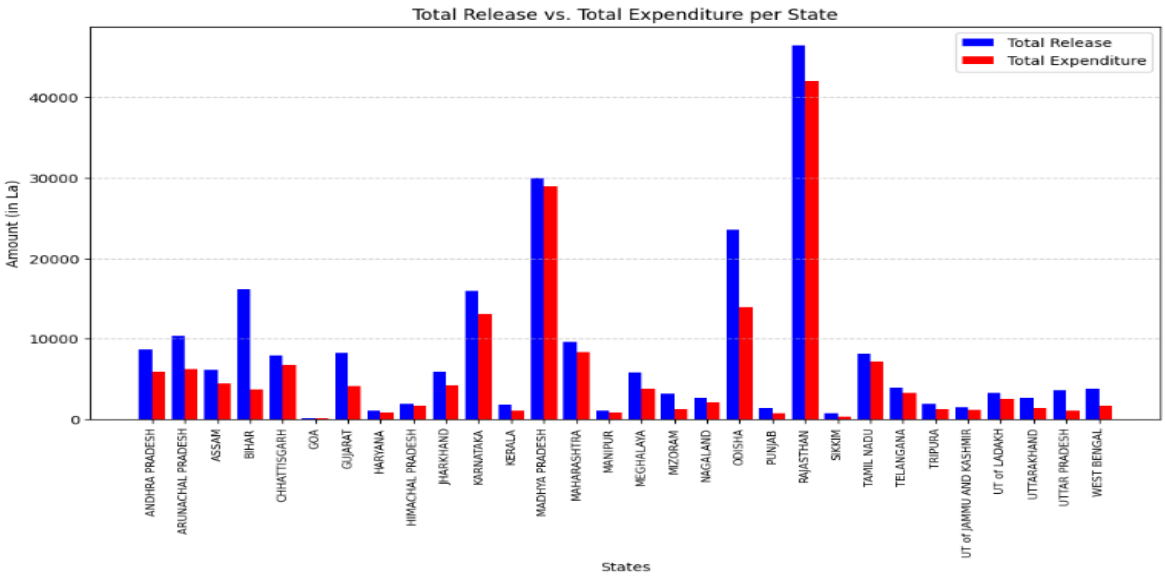


Figure 5.3. 1 Total Release vs Total Expenditure

Figure 5.3.2 uses Elbow Method for Optimal Clusters:

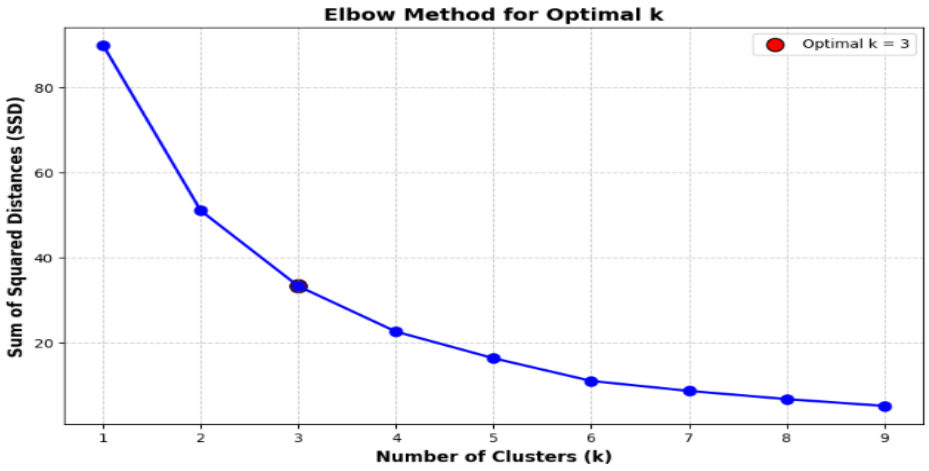


Figure 5.3. 2 Elbow Method for Optimal Clusters

Clustering States: States were clustered based on the calculated efficiency metrices i.e. Release Efficiency, Expenditure Efficiency and Overall Utilization. As shown in Figure 3.3.3

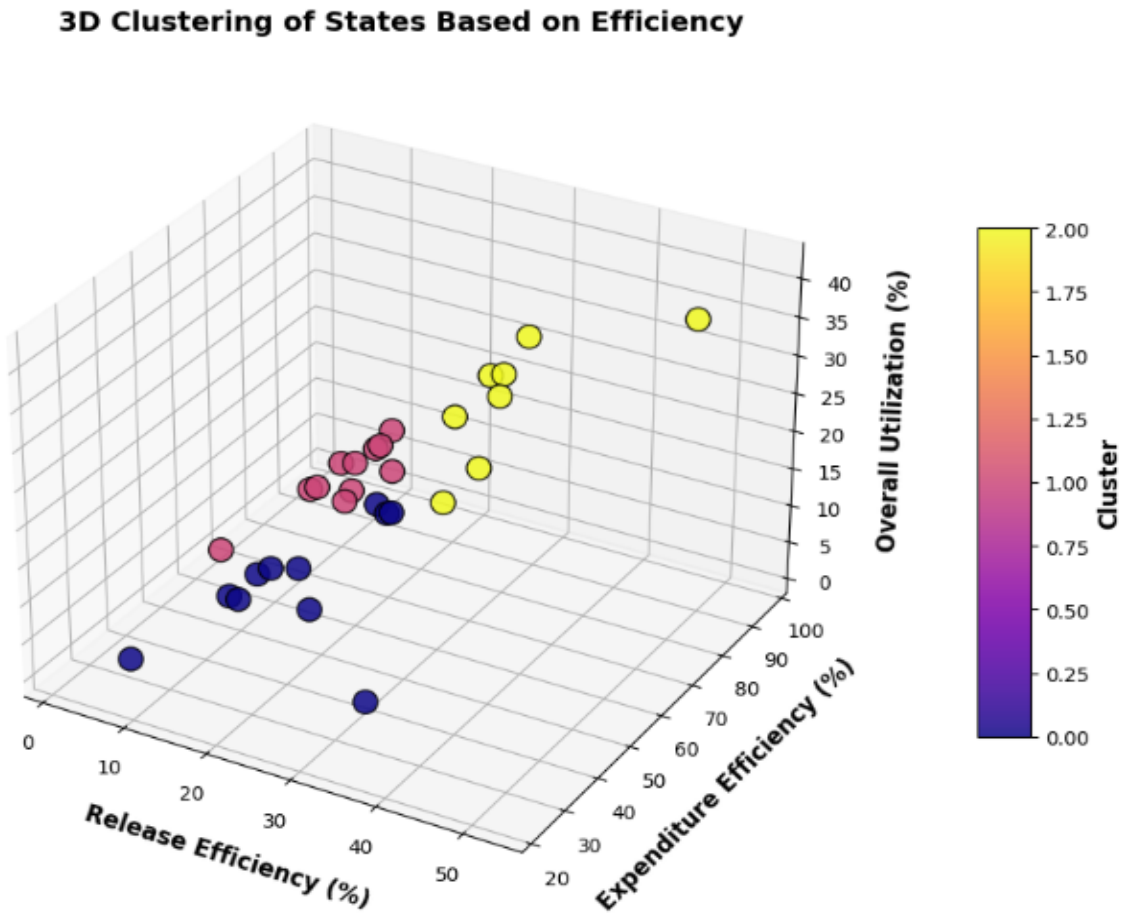


Figure 5.3. 3 CLustering of States

Inferences from Figure 5.3.3 are shown in Figure 5.3.4

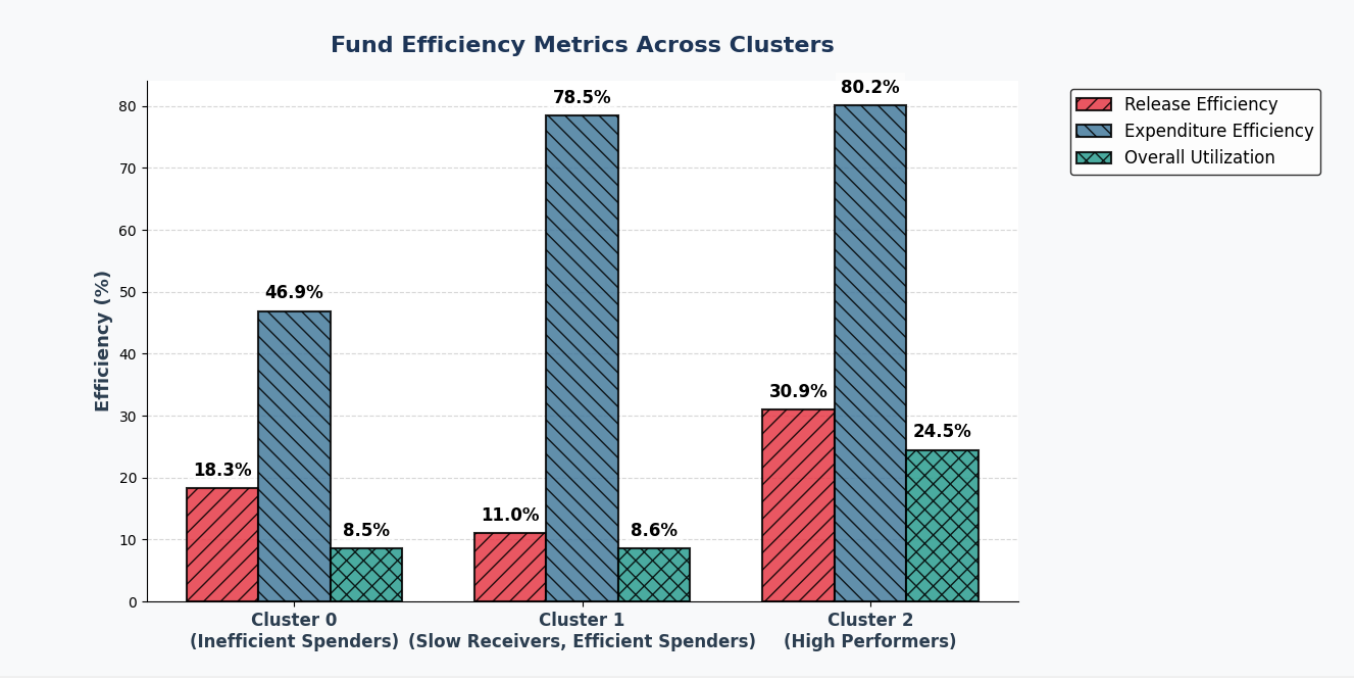


Figure 5.3. 4 Cluster Analysis

Table 5.3 shows cluster differentiation.

Cluster 0 → "Inefficient Spenders "having 11 States	**Cluster 1 →"Slow Receivers, Efficient Spenders "having 11 States**	**Cluster 2 →"High Performers "having 8 States**
Moderate fund release but poor spending and utilization. Struggles to effectively use allocated funds. Moderate Release Efficiency (18.27%) Low Expenditure Efficiency (46.91%) Low Overall Utilization (8.51%) These states receive funds at a moderate rate but struggle with spending efficiency and overall utilization.	Funds arrive slowly but are spent well. Utilization remains low due to slow fund release. Low Release Efficiency (10.98%) High Expenditure Efficiency (78.51%) Low Overall Utilization (8.63%) These states receive funds slowly but spend efficiently when they do. However, their overall utilization remains low due to slow fund release.	Best in all metrics—funds are released fast and used efficiently. These states maximize resource utilization. High Release Efficiency (30.94%) High Expenditure Efficiency (80.16%) High Overall Utilization (24.49%) These states are the best performers in all metrics. They get funds quickly and utilize them efficiently.

Table 5.3 Cluster Analysis

CHAPTER 6
OBJECTIVE 3

6.1 Discussion

Objective: Predicting Cost Per Hectare for Watershed Projects: Forecasting Funding Efficiency Based on Location and Investment.

Predicting the cost per hectare is crucial for efficient budget planning in watershed projects. Using a Random Forest Regressor, the model analyzed key financial and geographical factors such as total sanctioned cost**,** central and state share**,** and area sanctioned to estimate costs for new projects as shown in Table 6.1. The model achieved a strong R^2 score of 0.88, indicating high predictive accuracy. These insights can help policymakers optimize fund allocation, ensuring resources are distributed effectively based on project needs and regional variations.

Features	Target
Central Share	Cost_Per_Hectare
State Share	
Central_Funding_Ratio	
State_Funding_Ratio	
State_Avg_Area	
District_Avg_Cost	
Area_Central_Interaction	
Central_State_Ratio	
Log_Central_Share	
State_Name_Encoded	
District_Name_Encoded	

Table 6.1 Features & Target

6.2 Implementation

- **Adding New Features to Improve Data**

  Newly calculated features like cost per hectare, funding ratios, state & district averages were incorporated in model training to increase model accuracy.

- **Changing Categorical Information**

  Label encoding was performed on State and district names so the model could process them effectively.

- **Getting Training Data Ready**

  Multiple features as mentioned in *Table 6.1* were used to train model and find their relationship/pattern with the target variable i.e. cost per hectare.

- **Model Training and Evaluating**

  Model was trained using random forest regressor as the goal was to predict a continuous value. The model was then evaluated using MSE, RMSE and R^2 scores.

- **User Input to Make Predictions**

  A functioned was developed allowing users to directly enter details of new projects like State name, district name, area sanctioned, central share and state share from the terminal.

6.3 Results

The model performed well and was successfully able to predict cost per hectare of new projects with the following evaluation metricsas shown in Figure 6.3.1

```
Enter details for a new project:
State Name: ANDHRA PRADESH
District Name: ANANTAPUR
Central Share: 570
State Share: 377
Area Sanctioned: 3360

Prediction Results:
Predicted Cost Per Hectare: 0.2704
Estimated Total Cost for 3360.0 hectares: 908.54

Mean Squared Error (MSE): 0.000108
Root Mean Squared Error (RMSE): 0.010371
R-squared (R²): 0.8804
```

Figure 6.3. 1 Cost_Per_Hectare Prediction

CHAPTER 7
EXPERIMENT AND RESULT ANALYSIS

7.1 Findings and Results

Beneficiary analysis revealed the way watershed projects helped farmers and created employment opportunities within the industry. The report highlighted that regions were facing difficulty in implementing their expenditures which led to uneven results in project implementation, while others were able to create significant mandays and help many farmers. These variations were even more evident through the clustering study that divided the regions on the basis of their efficiency.

Moreover, the cost per hectare prediction model yielded accurate estimates with R2 of 0.88, which would enable proper allocation of funds. With such information, government and policymakers can make more informed fund allocation decisions and target projects that are likely to yield maximum outcomes.

7.2 Conclusion

The research cast significant illumination on the influence of watershed projects, how stakeholders are impacted, and the economics that propel success or failure. The data considered indicated that farmers gained numerous favorable advantages, mandays were generated, and money was invested which helped identify other areas where improvement was needed. The estimated cost per hectare provided a grassroots foundation for improved fiscal management, while the model of clustering revealed inequalities in the effectiveness of fund use. Such results can contribute to making interventions in the future more effective, simple, and less wasteful in rural areas.

7.3 Objectives Achieved

- Beneficiary Assessment of Watershed Projects: Analyzing Budget, Execution, Mandays Generated and Farmers Benefitted.
- Fund Flow Efficiency Analysis of Watershed Projects in India.
- Predicting Cost Per Hectare for Watershed Projects: Forecasting Funding Efficiency Based on Location and Investment.

7.4 Limitations and Future Enhancements

Study Limitations:

Although the research was informative, there were some limitations. Because the analysis was based on historical information, it is possible that it may not fully reflect how recent policy adjustments or unforeseen economic changes affect the use of funds. Moreover, differences in the consistency and availability of data between states could have resulted in an incomplete comprehension of regional disparities. Another key consideration is that the model's projections are made from historical trends, so accuracy may be impacted new changes.

Future Enhancements:

Future development may also emphasize bringing in real-time data to improve forecast accuracy and applicability. Bringing in satellite images and climate data may provide more insights into how weather, landscape, and other environmental variables influence watershed project performance. Enhancing the clustering model through more granular state or district-level data may further enhance fund allocation strategies. Creating an interactive policymaker dashboard would also make it simpler to access and utilize data-driven decision-making.

7.5 Summary

India's watershed project evaluation system is structured to give straightforward and significant views of project impact, expenditure efficiency, and allocation of funds. It consolidates data on budgeted expenditures, treasury fund disbursements, actual spending, and their consequences—like the number of farmers benefited and job opportunities generated. The process is a systematic one, with data pre-processing through VS Code, analysis and visualization through Python Colab, and interactive geographic maps through Power BI.

While the system enhances decision-making and transparency, there are some challenges. Inconsistent data from multiple sources, delayed updates, and challenges in connecting financial inputs to real-world outcomes can impact accuracy. These challenges will need to be addressed through strong data cleaning, statistical validation, and effective visualization methods to ensure that the insights produced are both reliable and actionable.

7.6 References

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